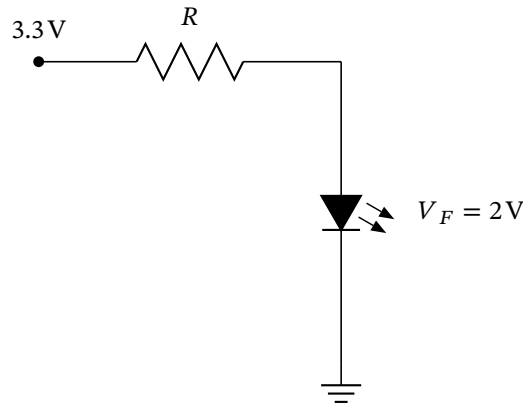


Current-Limiting Resistor for 7-Segment Display

To determine an appropriate current-limiting resistor for the 7-segment display, we select a target of 8 mA passing through each segment when it is lit, so that it is within the provided constraints. We note a typical forward voltage is $V_F = 2\text{ V}$ from the datasheet, so the circuit for a single segment LED looks something like:



Using Ohm's law, the resistance must be:

$$\frac{3.3\text{ V} - 2\text{ V}}{R} = 8\text{ mA}$$
$$R = 162.5\ \Omega$$

This is not a standard resistance value, so we will pick the closest value of $150\ \Omega$ instead. This results in a current draw of 8.667 mA instead, which is still within spec.

LED Blink Frequency

Note: the spec sheet indicates a target frequency of 2 Hz but the actual lab instructions say to target a frequency of 2.4 Hz instead. I will be using the latter.

The HSOSC module connects to an on-board high-frequency oscillator at 48 MHz. One period of the clock cycle is therefore 20.833 ns. We wish to blink the LED at 2.4 Hz, which has a period of 0.417 s.

My `clk_freq_divider` verilog module toggles the LED on/off each time its counter reaches the MAXCOUNT value. Thus, we should set MAXCOUNT so that the counter takes half of the desired period to overflow back to zero (and thus a single on/off cycle will sum to one whole period):

$$20.833\text{ ns} \cdot \text{count} = \frac{0.417\text{ s}}{2}$$
$$\text{count} = 10000000$$

Thus, we will set MAXCOUNT to a value of 10000000 to achieve the desired blinking frequency.